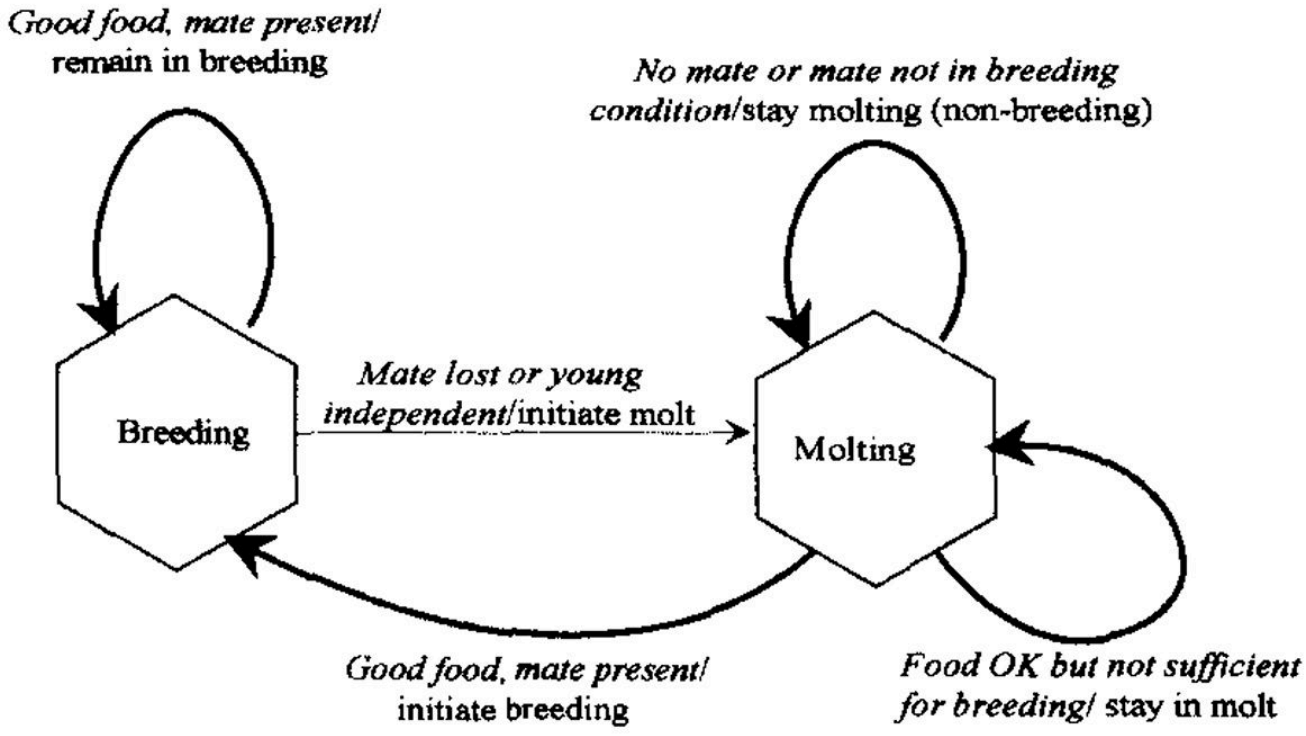


Annual Cycles

Dr. Ethan Linck
Office: 305 Lewis Hall

Resident life cycles

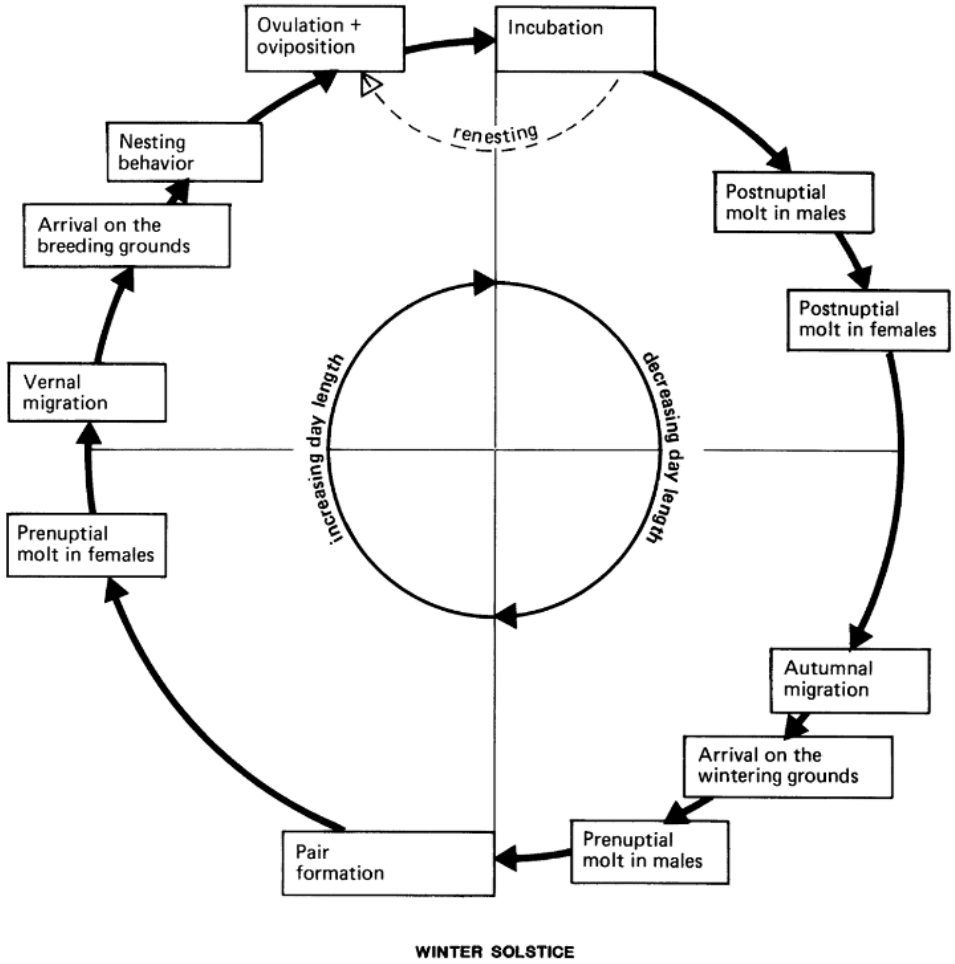
- Three tasks: breed, molt, survive
- Feral pigeons cycle between breeding and molting (the nonbreeding stage) based on presence of mate, presence of food, and development of young



(Jacobs & Wingfield 2000)

Migratory life cycles

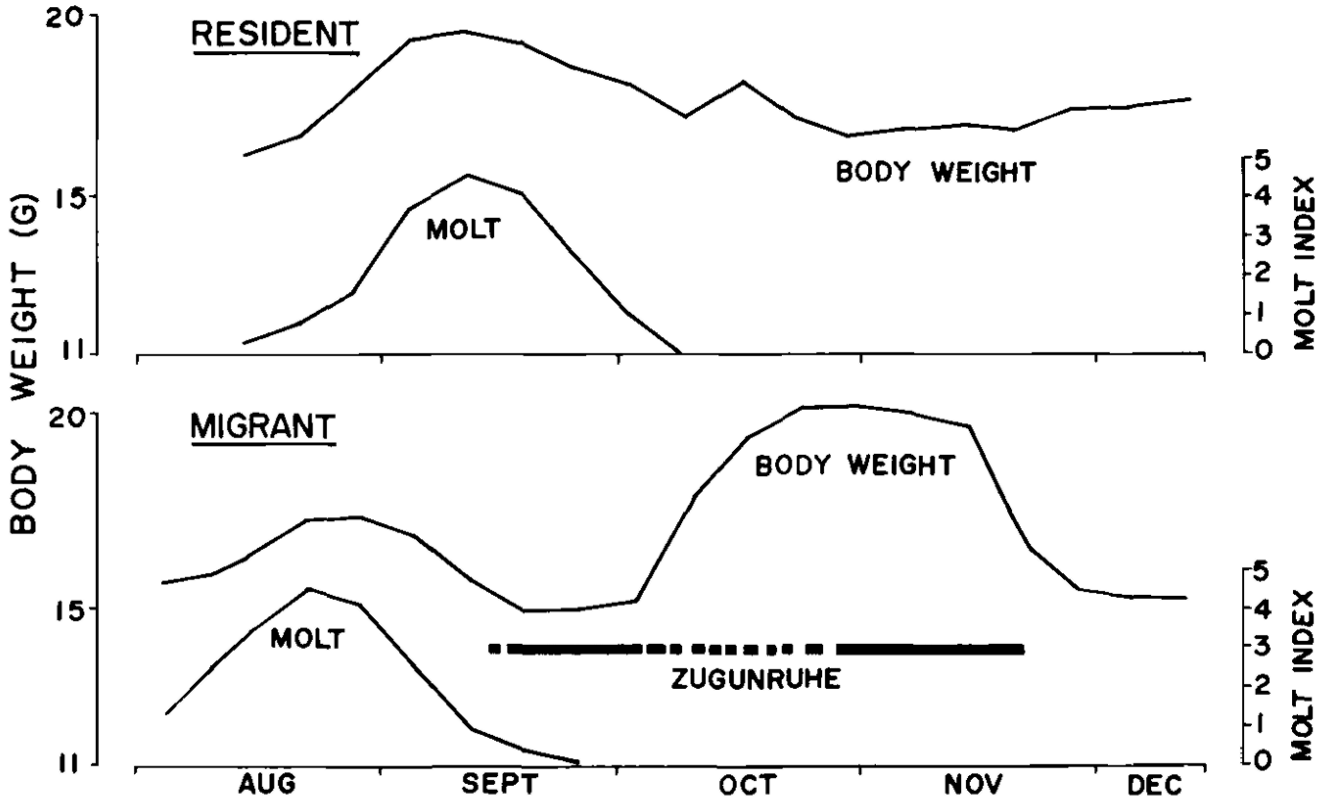
- Four tasks: breed, molt, migrate, survive
- Mediated by **photoperiod** (day length)



(Bluhm 1988)

Molt and migration are energetically costly

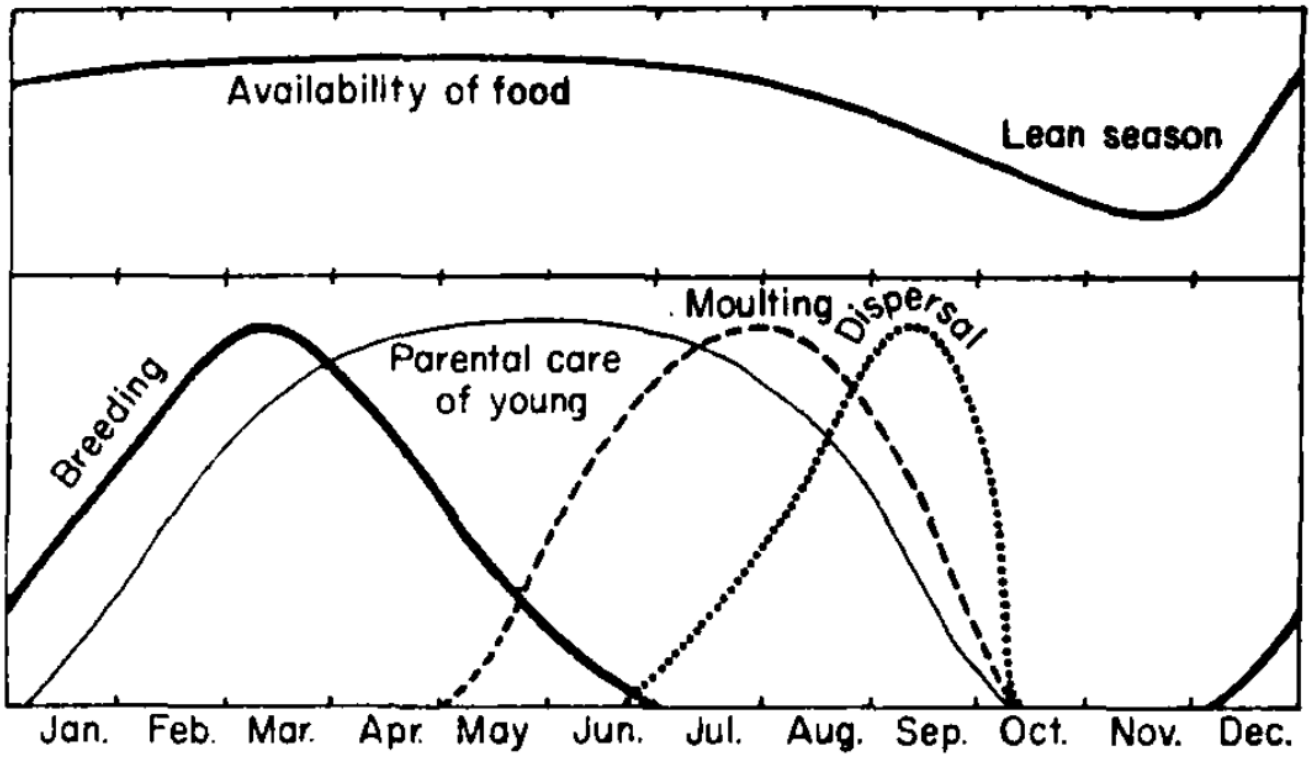
- Migratory European Robins molt earlier, have two body mass peaks (during molt and **zugunruhe** migratory restlessness period)
- Residents molt later (= one body mass peak)



(Biebach 1983)

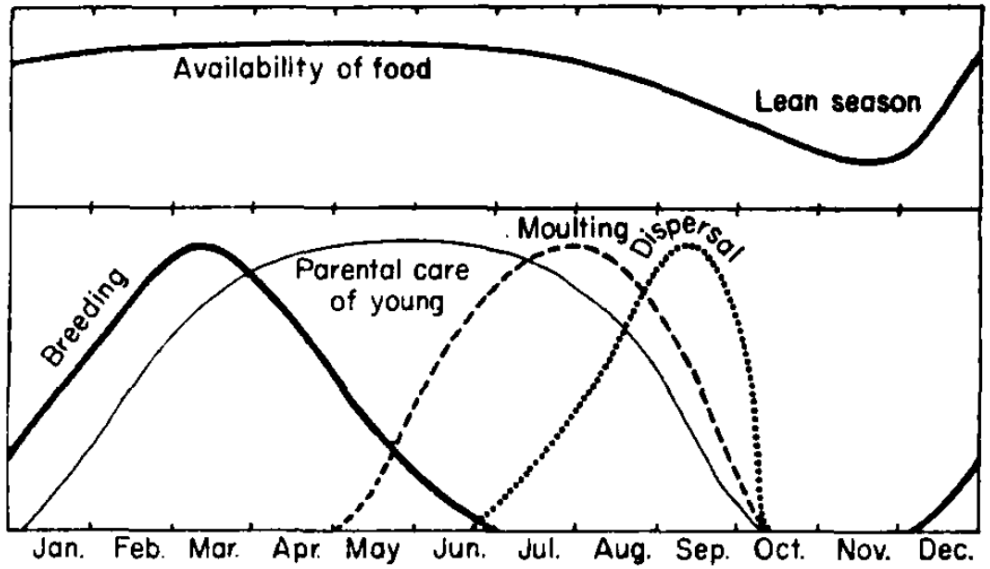
Breeding and molting coincide with period greatest food availability

- Tropical environments: driven by seasonal variation in rainfall
- Temperate environments: driven seasonal variation in temperature



(Fogden 1972)

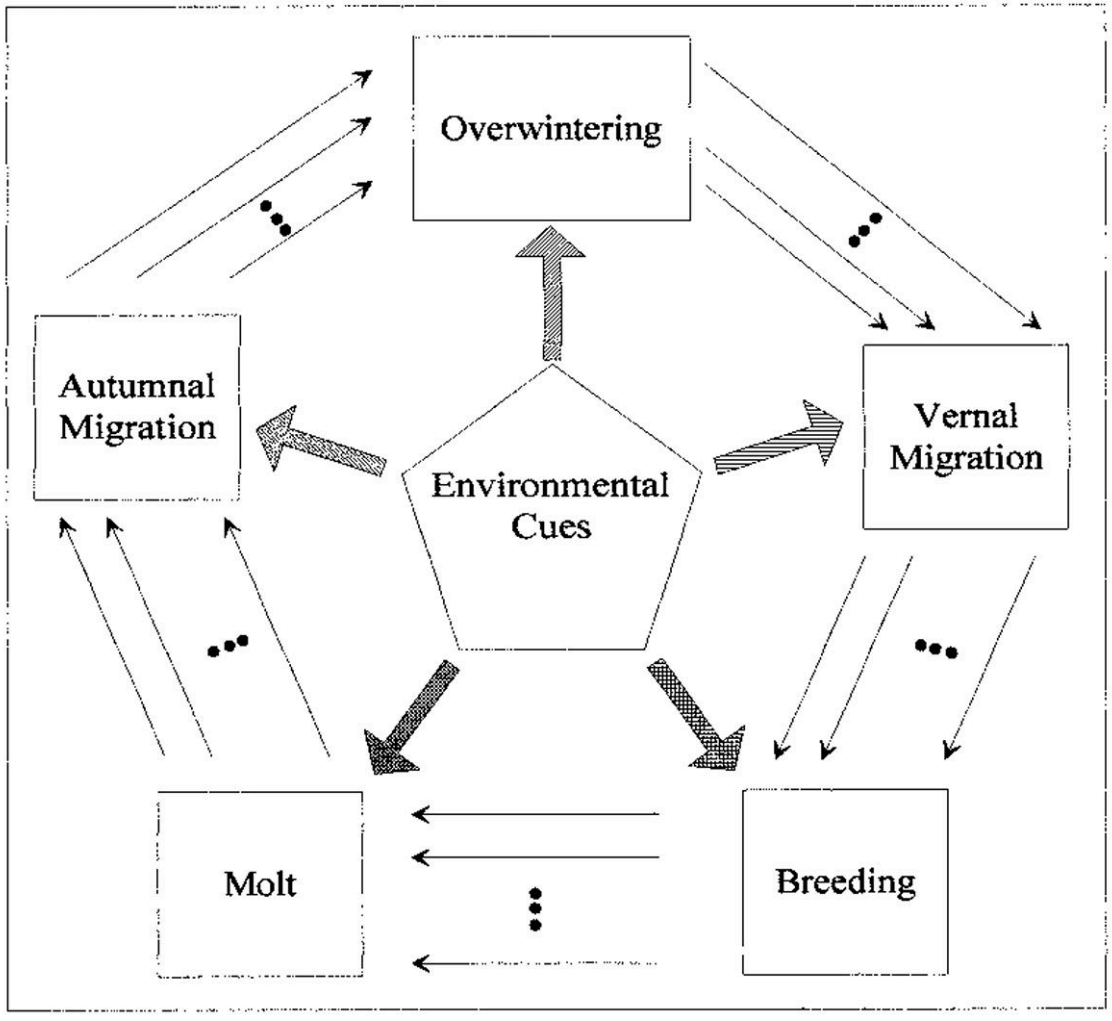
When would we expect body mass peaks in tropical birds?



a) molting
b) breeding
c) dispersal
d) "the lean season"
e) a, b, or c

(Fogden 1972)

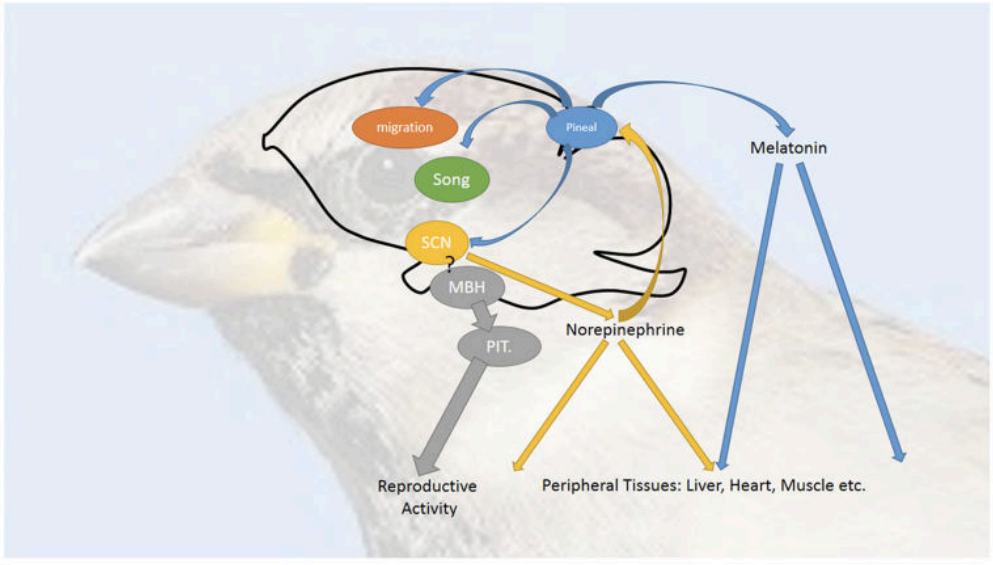
Life cycle stages depend on physiological state and environmental cues



(Jacobs & Wingfield 2000)

Biological clocks are controlled by pineal gland

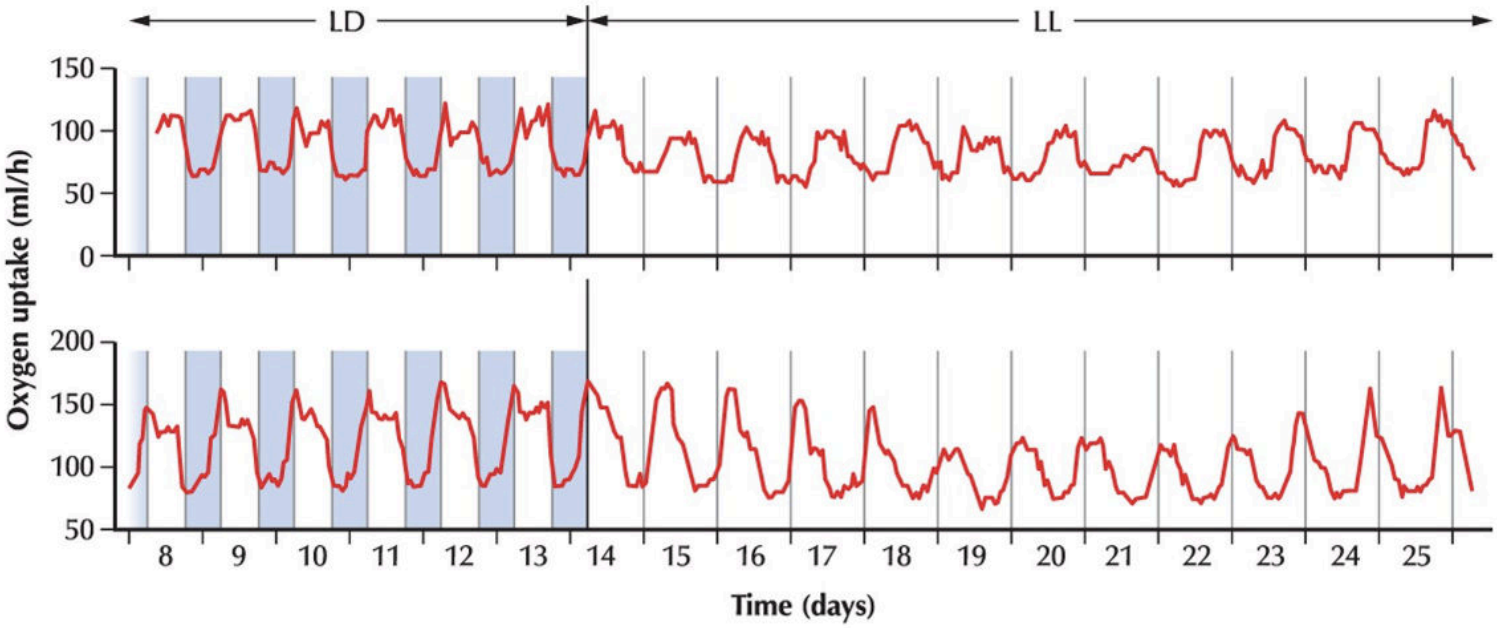
- Located on top of brain
- **Circadian rhythms** (24 hour cycles) influenced by **pineal gland** (releases **melatonin** into blood), **suprachiasmatic nuclei** in hypothalmus (regulates metabolism), and eye (light recievers)
- **Circannual rhythms** (1 year cycles) influenced by photoperiod



(Cassone 2014)

Instrinsic circadian rhythms are imperfect

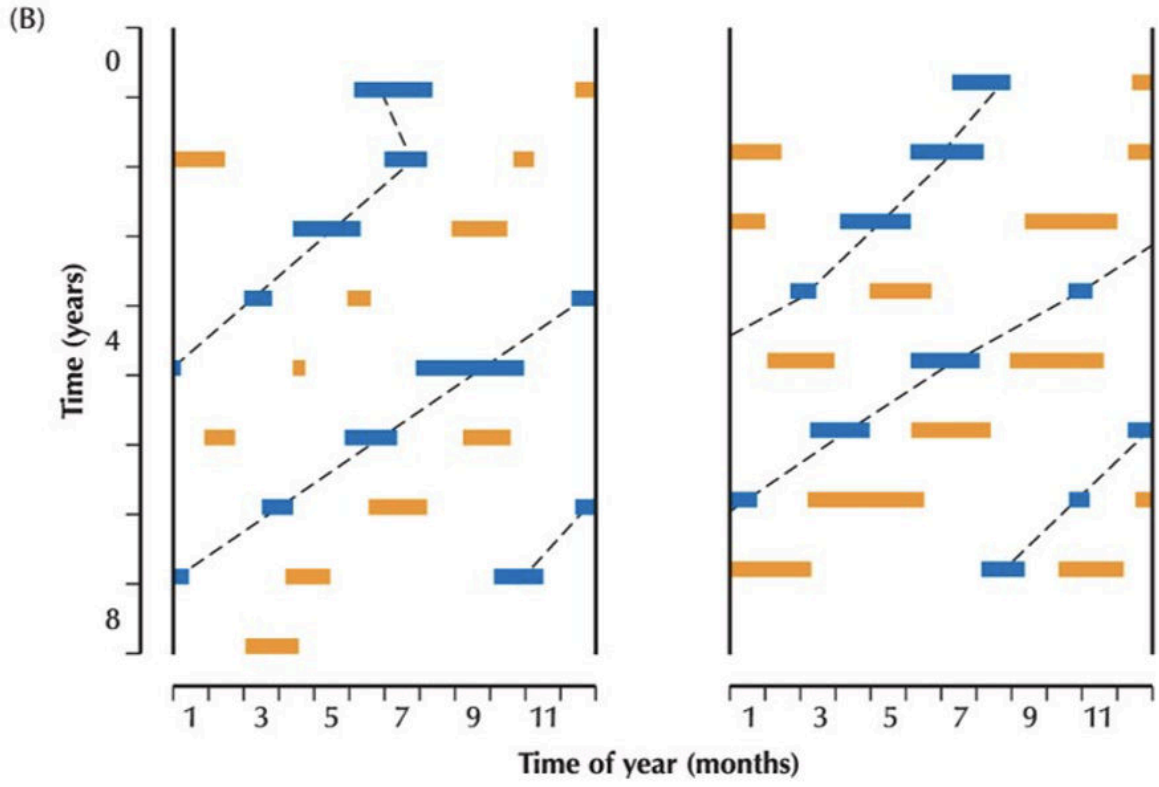
- 24 hour clock maintained by **zeitgebers** (environmental cues)
- Daily activity cycles (measured by metabolism) drift in chaffinches



(Gill et al. 2019)

Instrinsic circannual rhythms are, too

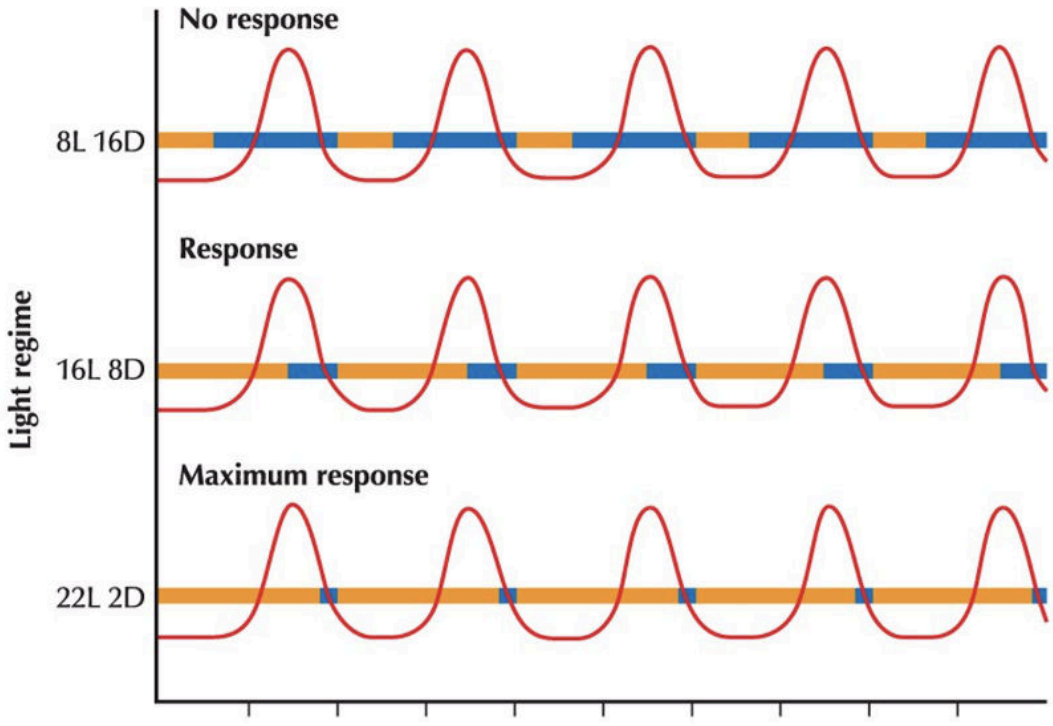
- Summer molt (blue) and winter molt (orange) drift earlier in Garden Warblers (L) and Eurasian Blackcaps (R) kept in captivity for eight years
- Consistent with intrinsic 10 month cycle



(Gill et al. 2019)

Photoperiod measured by overlap of day length and peak activity

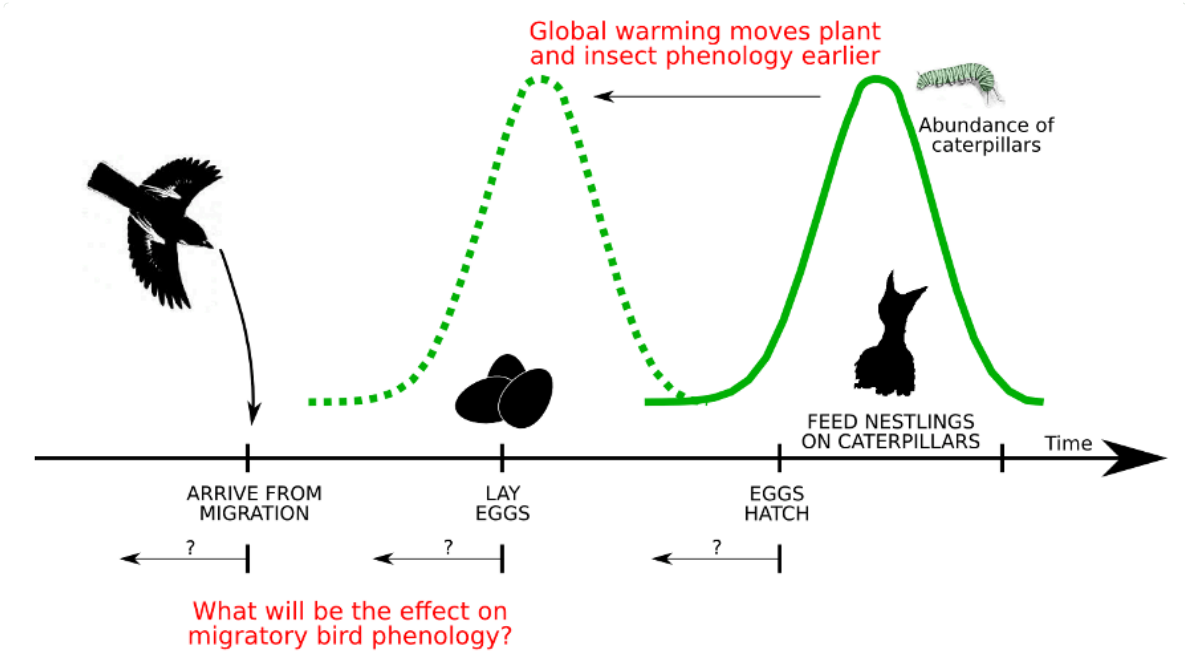
- Known as the **external coincidence model**
- "Response" measurement gonad size—why?



(Gill et al. 2019)

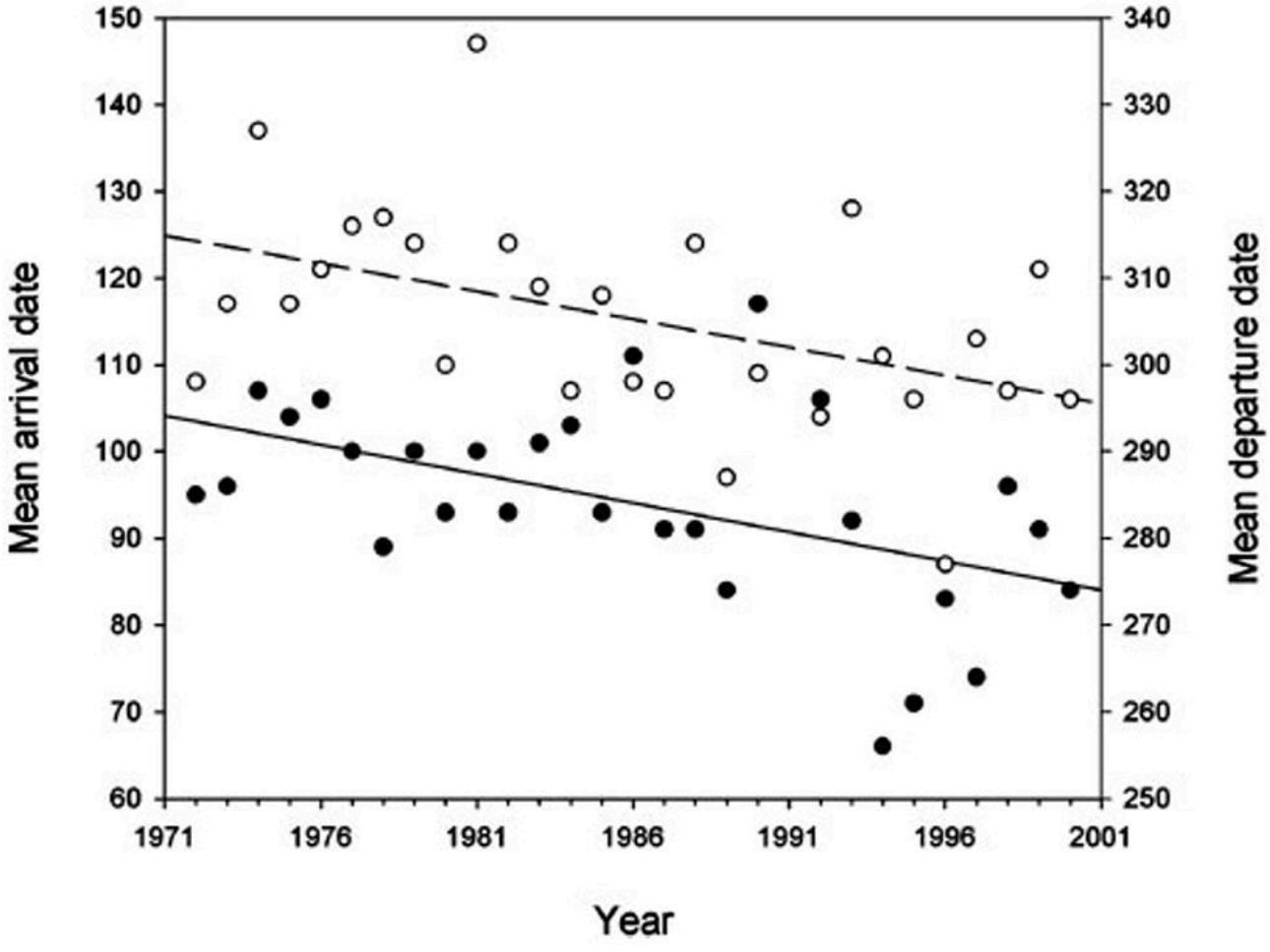
Phenological mismatch and climate change

- **Phenological mismatches** occur when the timing of life cycle events of interacting species no longer aligns



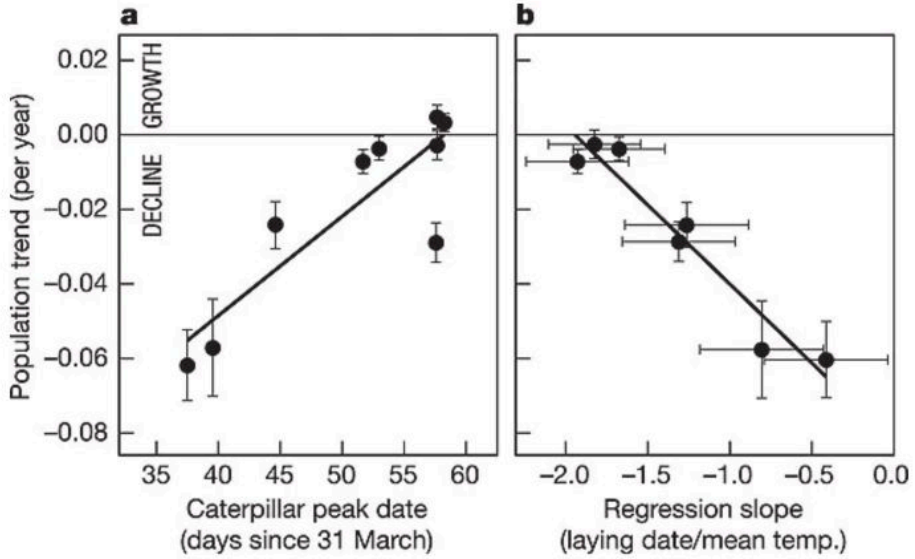
N. Kristensen

Arrival dates moving for some species



(Cotton 2003)

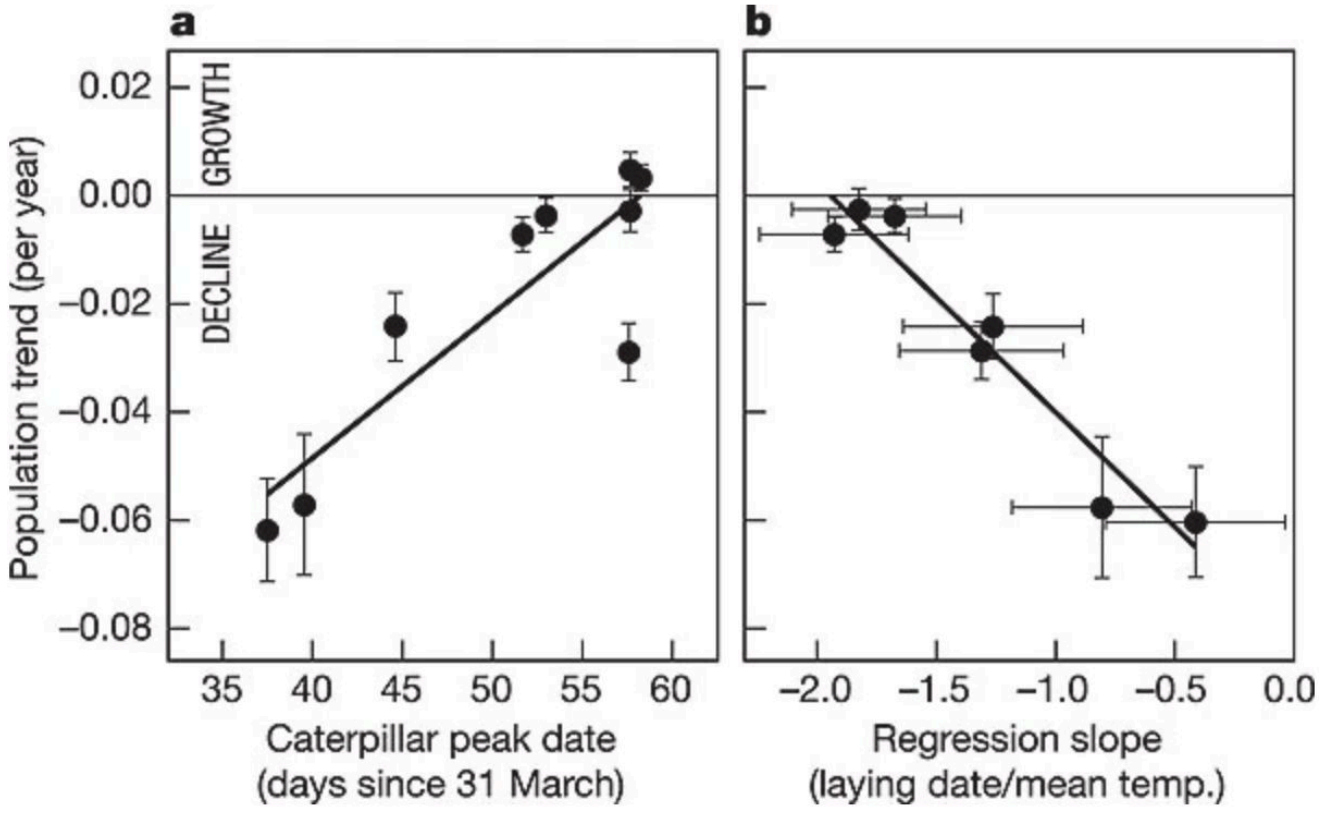
What can you conclude about populations of Dutch Pied Flycatchers?



a) earlier peak caterpillar abundance leads to population growth
b) populations that lay eggs much earlier when it is hot decline more strongly
c) populations that do not lay eggs much earlier when it is hot decline more strongly
d) a and b
e) none of the above

(Both et al. 2019)

Population declines when arrival date fails to track food resources



(Both et al. 2019)

References

- Title slide species: *Sibirionetta formosa*, *Sylviorthorhynchus desmursii*, *Picumnus varzeae*, *Ficedula subrubra*
- Bluhm, C. K. (1988). Temporal patterns of pair formation and reproduction in annual cycles and associated endocrinology in waterfowl. Current ornithology, 123-185.
- Jacobs, J. D., & Wingfield, J. C. (2000). Endocrine control of life-cycle stages: a constraint on response to the environment?. The Condor: Ornithological Applications, 102(1), 35-51.
- Biebach, H. (1983). Genetic determination of partial migration in the European robin (Erithacus rubecula). The Auk, 100(3), 601-606.
- Fogden, M. P. L. (1972). The seasonality and population dynamics of equatorial forest birds in Sarawak. Ibis, 114(3), 307-343.

References

- Cassone, V. M. (2014). Avian circadian organization: a chorus of clocks. Frontiers in neuroendocrinology, 35(1), 76-88.
- Gill, F. B., Prum, R. O., & Robinson, S. K. (2019). Ornithology (4th ed.). W. H. Freeman.
- Cotton, P. A. (2003). Avian migration phenology and global climate change. Proceedings of the National Academy of Sciences, 100(21), 12219-12222.
- Both, C., Bouwhuis, S., Lessells, C. M., & Visser, M. E. (2006). Climate change and population declines in a long-distance migratory bird. Nature, 441(7089), 81-83.